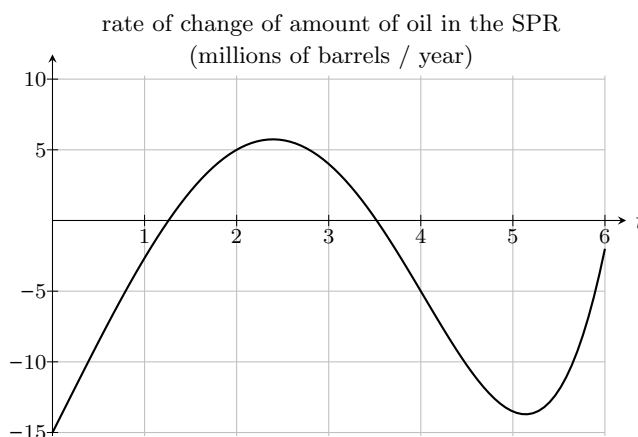


Net Change/ The Definite Integral

1

The Strategic Petroleum Reserve (SPR) is an emergency supply of oil maintained by the United States Department of Energy. The graph below is a model of $f(t)$, the *rate* at which oil enters or leaves the reserve in millions of barrels per year (a positive rate means oil is entering; a negative rate means oil is leaving). Time t is measured in years.



We'd like to approximate the net change in the amount of oil in the SPR between $t = 0$ and $t = 6$. As you learned in class, the idea is to pretend that the rate of change is constant for short time intervals.

- (a) At what rate was oil entering/leaving the SPR at $t = 0$? (For this part and all following, give your answer *with units*, and be sure to say whether the oil was entering or leaving.)

Solution

Since $f(0) = -15$, oil was leaving at a rate of 15 million barrels per year.

- (b) If we *pretended*, for the purposes of simplicity, that oil entered/left the reserve at a *constant* rate (namely, the rate you found in (a)) between $t = 0$ and $t = 2$, how much oil entered/left in that time period?

Solution

We're pretending that oil left at a constant rate of 15 million barrels per year for 2 years; in that case, $(15 \text{ million barrels/year})(2 \text{ years}) = 30 \text{ million barrels}$ would have left the reserve.

- (c) At what rate was oil entering/leaving the reserve at $t = 2$?

Solution

Since $f(2) = 5$, oil was entering at a rate of 5 million barrels per year.

- (d) If we pretended that oil entered/left the reserve at this same rate between $t = 2$ and $t = 4$, how much oil entered/left in that time period?

Solution

Then, (5 million barrels/year)(2 years) = 10 million barrels would have entered.

- (e) At what rate is oil entering/leaving the reserve at $t = 4$?

Solution

Since $f(4) = -5$, oil was leaving at a rate of 5 million barrels per year.

- (f) If we pretended that oil entered/left the reserve at this same rate between $t = 4$ and $t = 6$, how much oil entered/left in that time period?

Solution

Then, (5 million barrels/year)(2 years) = 10 million barrels would have left.

- (g) Use your answers to (b), (d), and (f) to estimate the net change in the amount of oil from $t = 0$ to $t = 6$. Was this a net gain or a net loss?

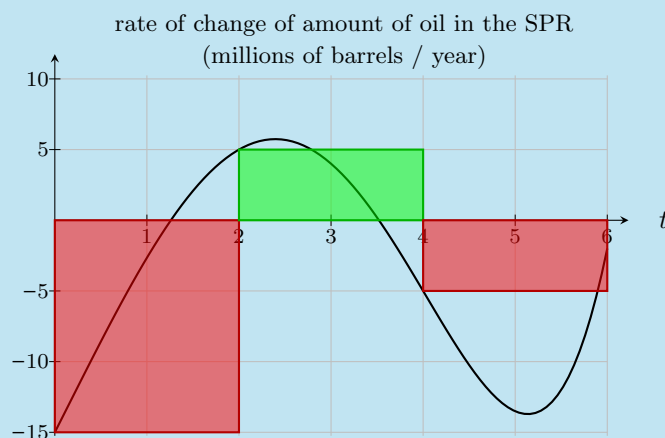
Solution

We estimated that the reserve would lose 30 million barrels, gain 10 million barrels, then lose another 10 million barrels, for a net change of $-30 + 10 - 10 = -30$ million barrels. This is a net loss of 30 million barrels.

- (h) Illustrate your answer to (g) on the graph of f . Did you do a right-hand or left-hand approximation?

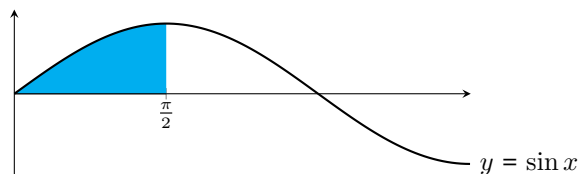
Solution

The net change can be represented as a signed area of rectangles, with the green area contributing a positive net change and the red area contributing a negative net change:



2

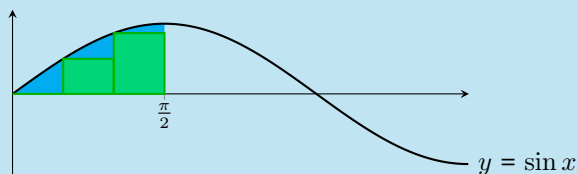
In this problem, we are interested in the area between the graph of $\sin x$ and the x -axis for x in the interval $\left[0, \frac{\pi}{2}\right]$. (This is the shaded area in the graph below.)



- (a) Recall that we use the notation L_3 to mean the left-hand sum with three rectangles (of equal width) that estimates this area. Find L_3 ; give an exact answer. Draw a picture illustrating L_3 . Is L_3 an overestimate or underestimate for the area we're interested in?

Solution

We can picture L_3 as the area in the green rectangles, which shows that it's an underestimate.



(Note that the first rectangle has height 0.)

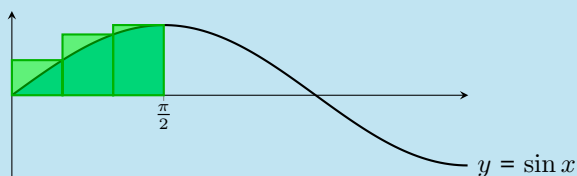
Each of the rectangles has width $\frac{\pi}{6}$, so

$$\begin{aligned} L_3 &= \sin(0) \cdot \frac{\pi}{6} + \sin\left(\frac{\pi}{6}\right) \cdot \frac{\pi}{6} + \sin\left(\frac{2\pi}{6}\right) \cdot \frac{\pi}{6} \\ &= 0 + \frac{1}{2} \cdot \frac{\pi}{6} + \frac{\sqrt{3}}{2} \cdot \frac{\pi}{6} \\ &= \frac{\pi(1 + \sqrt{3})}{12} \end{aligned}$$

- (b) Repeat (a) using a right-hand sum rather than a left-hand sum.

Solution

This time we are interested in R_3 , which is an overestimate:



$$\begin{aligned} R_3 &= \sin\left(\frac{\pi}{6}\right) \cdot \frac{\pi}{6} + \sin\left(\frac{2\pi}{6}\right) \cdot \frac{\pi}{6} + \sin\left(\frac{3\pi}{6}\right) \cdot \frac{\pi}{6} \\ &= \frac{1}{2} \cdot \frac{\pi}{6} + \frac{\sqrt{3}}{2} \cdot \frac{\pi}{6} + 1 \cdot \frac{\pi}{6} \\ &= \frac{\pi(3 + \sqrt{3})}{12} \end{aligned}$$

- (c) Use the Riemann Sums¹ applet at <https://www.geogebra.org/m/MYn6MgTQ> to find L_{50} and R_{50} . (You can also use it to check your answers for L_3 and R_3 .) Based on these, what do you think the actual area we're looking for might be?

To use the applet, type $\sin(x)$ in the box for $f(x)$, 0 in the box for a , and $\pi/2$ in the box for b ; and $n = 50$. This will give you a left-hand sum, a right-hand sum, and something called a midpoint sum which is (not surprisingly) somewhere in between.

Solution

L_{50} is 0.98421 and R_{50} is 1.01563, which suggests the exact area might be 1.

- (d) When you estimate the area under the curve $y = \sin(x)$ between 0 and $\frac{\pi}{2}$ using 50 rectangles, what is the width of each rectangle in your figure?

Solution

We are breaking up the interval between $x = 0$ and $x = \frac{\pi}{2}$ into 50 pieces, so each piece is $\frac{\pi}{100}$ wide.

- (e) When you estimate the area under the curve $y = \sin(x)$ between 0 and $\frac{\pi}{2}$ using n rectangles, what is the width of each rectangle in your figure? Give your answer in terms of n and justify your reasoning.

Solution

We are breaking up the interval between $x = 0$ and $x = \frac{\pi}{2}$ into n pieces, so each piece is $\frac{\pi}{2n}$ wide.

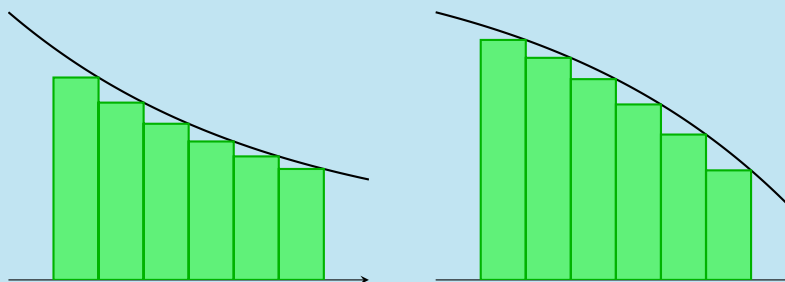
3

Reflect Back. What characteristic of f will ensure that a right-hand sum gives an underestimate of the signed area between the graph of f and the x -axis? Does it matter whether f is positive or negative? Increasing or decreasing? Concave up or concave down?

Illustrate your answer with some pictures.

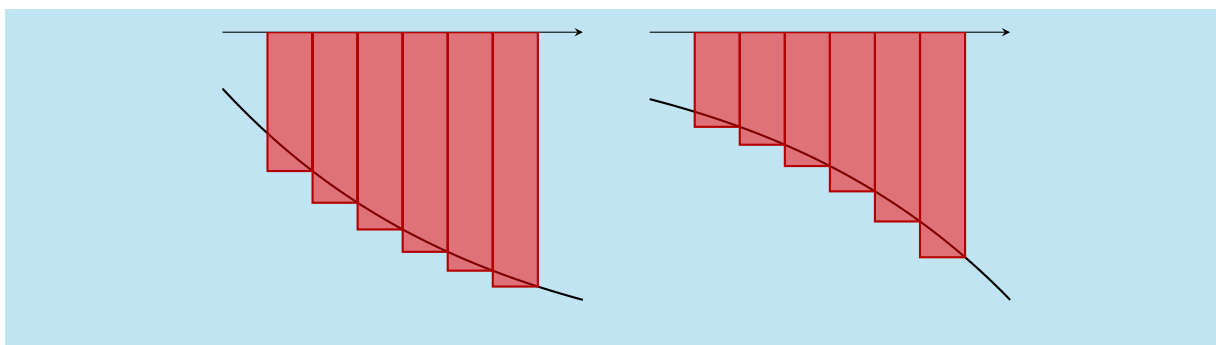
Solution

The function f needs to be decreasing for this to be true, but the concavity does not matter:



It's also true if the function is negative because the right-hand sum is then more negative than the actual signed area:

¹“Riemann sum” is just another way of saying “left-hand sum or right-hand sum”. These sums are named for Bernhard Riemann, a 19th century mathematician.



4 Evaluate the following definite integrals by interpreting each as a signed area. It will be very helpful to draw a graph.

(a) $\int_{-1}^3 3 \, dx$

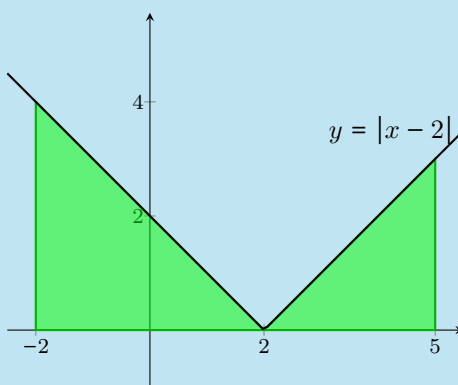
Solution

This is the signed area under $y = 3$ from $x = -1$ to $x = 3$. That region is a rectangle above the x -axis of height 3 and width $3 - (-1) = 4$, so the signed area is $\boxed{12}$.

(b) $\int_{-2}^5 |x - 2| \, dx$

Solution

We can picture this as the following signed area:

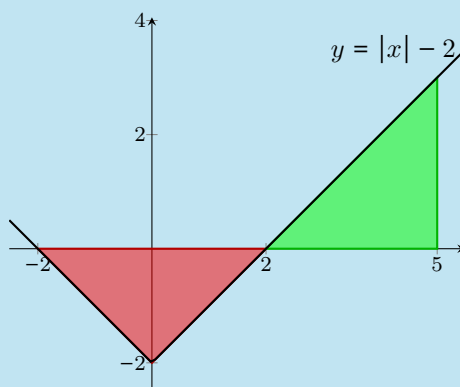


There are two triangles contributing to the area; one having base and height 4 (so area 8), the other having base and height 3 (so area $9/2$). The total area is $\boxed{\frac{25}{2}}$.

(c) $\int_{-2}^5 (|x| - 2) \, dx$

Solution

We can picture this as the following signed area:

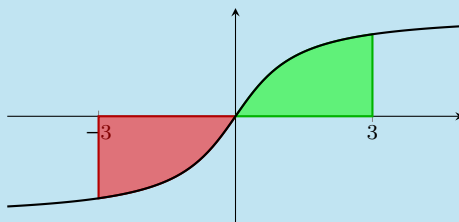


The green triangle has width and height 3, so its area is $\frac{9}{2}$. The red triangle has width 4 and height 2, so its area is 4. Therefore, the total signed area is $\frac{9}{2} - 4 = \boxed{\frac{1}{2}}$.

(d) $\int_{-3}^3 \arctan(x) dx$

Solution

We can picture this as the following signed area:



The green and red regions are identically shaped, so their areas cancel each other, and the total signed area is $\boxed{0}$.

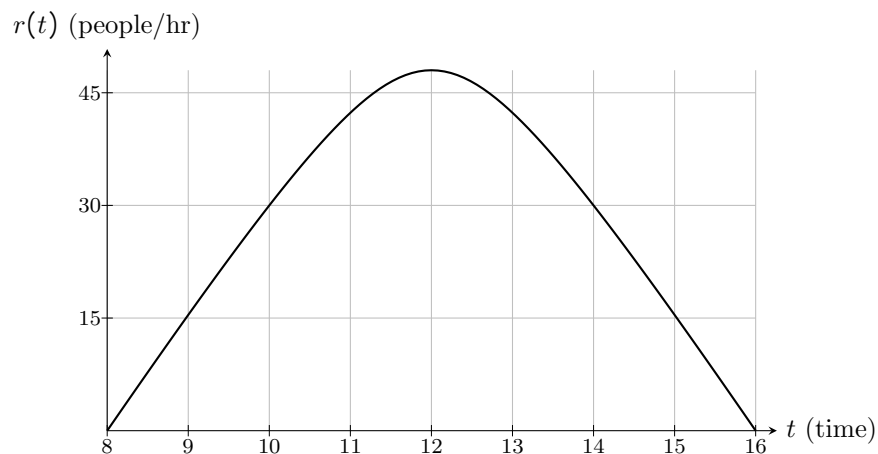
(e) $\int_{-3}^3 \sqrt{9 - x^2} dx$

Solution

$y = \sqrt{9 - x^2}$ is the top half of the circle of radius 3 centered at the origin, so this integral is the area of a half circle of radius 3, or $\frac{1}{2}\pi \cdot 3^2 = \boxed{\frac{9\pi}{2}}$.

5

A health clinic has set up a flu shot program for its patients. The clinic is open from 8 am to 4 pm (16:00) and gives flu shots on a first-come, first-serve basis. The clinic has the capacity to serve 30 patients per hour. The function $r(t)$, whose graph is given below, gives the rate at which people are arriving at the clinic for shots one day.



Explain your answers to the questions below by relating them to points, lengths, or areas on the graph.

- (a) If someone arrives at 9 AM, how long do they have to wait for a shot?

Solution

Because the clinic can serve 30 patients per hour and people are consistently arriving at less than 30 per hour, someone arriving at 9 AM should not have to wait at all to get a shot.

- (b) At what time does a line start forming?

Solution

This happens once people begin to arrive at the faster rate than the clinic can serve (30 people per hour), which is at 10 am.

- (c) At approximately what time is the length of the line increasing most rapidly?

Solution

This is when people are arriving at the fastest rate, which is at noon.

- (d) At approximately what time is the line the longest? (The answers to this is different from the answer to (c); explain why.)

Solution

Between 10 am and 2 pm, the length of the line is increasing; after 2 pm, the length of the line is decreasing (the rate at which people arrive is less than the rate at which they are served). Therefore, the line is longest at 2 pm ($t = 14$).

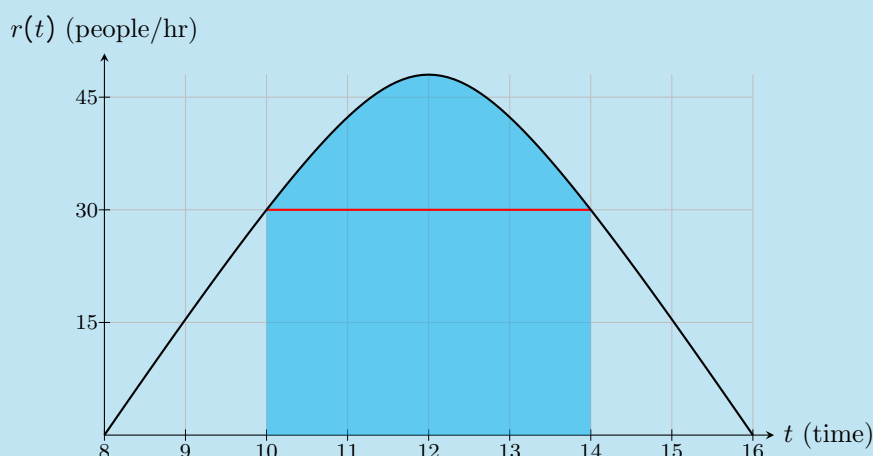
This differs from (c) because, between noon and 2 pm, people are arriving at a rate which is slower than the rate at noon but still higher than 30 per hour. This means the line will continue to lengthen, just not as quickly as at noon.

- (e) When the line is longest, approximately how many people are in line?

Solution

We're asked to find the length of the line at 2 pm ($t = 14$). Since the line started forming at $t = 10$, we can think of this as (number of people who arrived between $t = 10$ and $t = 14$) – (number of people who were served between $t = 10$ and $t = 14$). The number of people who arrived between $t = 10$ and $t = 14$ can be pictured as the area under $r(t)$ from $t = 10$ to $t = 14$ (the area marked in blue below); this area is about 165 (about 11 grid squares). The number of people who were served between $t = 10$ and $t = 14$ was $(30 \text{ people / hour})(4 \text{ hours}) = 120$, so there are approximately $165 - 120 = \boxed{45}$ people in line.

Here's an alternate way to picture it: the number of people who were served between $t = 10$ and $t = 14$ can be pictured as the area under $y = 30$ from $t = 10$ to $t = 14$ (marked in red below). Then, the length of the line is the area of the region which is just blue (not red), which is about 45 people (3 grid squares):



- (f) Approximate the longest amount of time a person could wait for a shot. (Assume that doors close to new arrivals at 4 pm but everyone who has arrived by 4 pm is served.)

Solution

The amount of time a person waits depends on how many people are in line before him. We already determined that there are about 45 people in line when the line is longest. A person arriving at this time has to wait for those 45 people to be served, which takes $\frac{45 \text{ people}}{30 \text{ people / hr}} = \boxed{1.5 \text{ hours}}$.

- (g) Approximately how long is the line at 3 pm?

Solution

We already figured out that the line has about 45 people in it at 2 pm, so let's figure out what happens between 2 and 3 pm. In this time, 30 people are served, but roughly 23 people arrive (this is the area under $r(t)$ from $t = 14$ to $t = 15$, which is about 1.5 grid squares). So, at 3 pm, the line has about $45 - 30 + 23 = \boxed{38 \text{ people}}$.

- (h) Approximate the number of people who came to the clinic for a flu shot this day.

Solution

This is the entire area under $r(t)$ from $t = 8$ to $t = 16$. This is about 15 grid squares in all, or about 225 people.

6

For the scenario described in #5, express the following quantities exactly; your answers may be in terms of definite integrals. (For instance, your answer might look like $50 + \int_8^{10} r(t) dt$ or $\int_8^9 [r(t) - 2] dt$.)

- (a) The length of the line at 11 am

Solution

Using the same reasoning as in #5(e), the length of the line at 11 am is (number of people who arrive between $t = 10$ and $t = 11$) – (number of people served between $t = 10$ and $t = 11$),

or $\int_{10}^{11} r(t) dt - 30$.

Solution

2 points. Give 1 point if they forget to subtract the number of people served.

- (b) The length of the line at 3 pm

Solution

The reasoning here is essentially the same as in (a); the number of people who arrive between 10 am and 3 pm is $\int_{10}^{15} r(t) dt$, while the number of people served in that time is (30 people /

hour)(5 hours) = 150. So, the length of the line at 3 pm is $\int_{10}^{15} r(t) dt - 150$.

Solution

2 points. Give 1 point if they forget to subtract the number of people served.

- (c) The amount of time a person arriving at noon has to wait in line

Solution

The length of the line at noon is $\int_{10}^{12} r(t) dt - 60$. A person arriving at noon has to wait for all of these people to be served, which happens at a rate of 30 people per hour. So, the waiting

time is $\frac{\int_{10}^{12} r(t) dt - 60 \text{ people}}{30 \text{ people/hour}}$ (the units simplify to hours).

- (d) The amount of time by which the clinic must extend its hours in order to serve everyone who arrives by 4pm

Solution

At 4 pm, the number of people in line is $\int_{10}^{16} r(t) dt - 180$. The amount of time it would take

to serve these people is $\boxed{\frac{\int_{10}^{16} r(t) dt - 180 \text{ people}}{30 \text{ people/hour}}}$.